

Rehearsal Reality: Exploring the Visualization of Experts' Mental Rehearsals from a First-Person Perspective to Support the Motor Learning of Novices

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Figure 1: *Left*: Learners can watch the mental rehearsal of an expert through a head mount display (HMD). *Right*: We conducted a four-day user study with between-subjects design to evaluate the impact of visualizing the mental rehearsals of experts on the motor learning of novices. Thirty-six novices were randomly assigned (12 per condition) to one of three conditions: positive video (PV), involving practice swings plus a first-person successful putting video; video mental rehearsal (VMR), involving practice swings plus a first-person video featuring experts' mental rehearsal; and XR mental rehearsal (XMR), involving practice swings while using XR apps that overlay experts' mental rehearsal onto the real world.

Abstract

Understanding the cognitive aspects of motor learning is crucial for novices to enhance learning and performance. One approach is mental rehearsal—"the cognitive rehearsal of a task prior to performance"—which pre-activates neural pathways and fosters deeper cognitive understanding. Although highly efficient, mental rehearsal can be challenging for novices, who often struggle to form vivid imagery of unfamiliar skills. We propose a concept of *rehearsal reality*, an XR-based application that overlays the mental rehearsals of experts onto real practice environment. We developed *rehearsal reality* for golf putting based on expert golfers' pre-putt mental imagery and conducted a qualitative study with 36 novice participants to understand novices' subjective learning experiences.

The results suggest that visualizing the mental rehearsals of experts can enhance the motor skills and external cognitive understanding of novices in golf putting. This finding can guide the potential of the mental rehearsals of experts as visual cues for motor learning.

CCS Concepts

• **Human-centered computing** → User studies; HCI theory, concepts and models; Mixed / augmented reality.

Keywords

Motor skill acquisition, Learning, Novices, Mental rehearsal, XR, Golf

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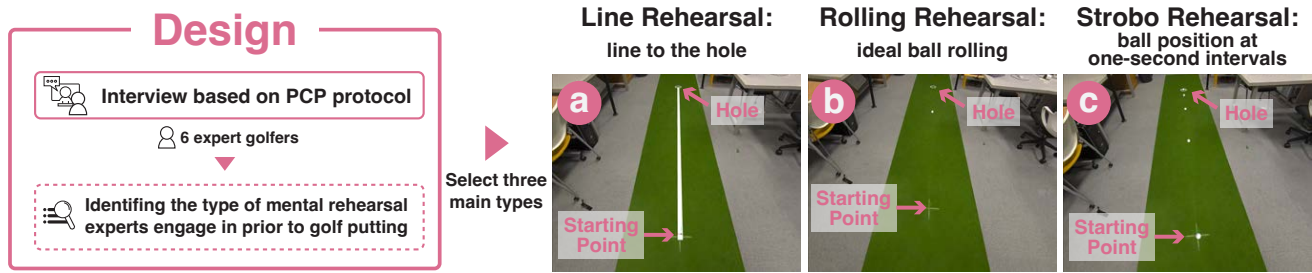


Figure 2: Process of developing *rehearsal reality*. First, online interviews were conducted with expert golfers, and reflexive thematic analysis was performed to identify the key aspects of their mental rehearsal that should be visualized. Based on this analysis, three mental rehearsals were visualized using XR technology, allowing them to be superimposed onto the real-world environment. (a) *Line rehearsal*: a white line from the ball to the cup; (b) *Rolling rehearsal*: an ideal rolling trajectory of the ball from its current position to the cup; (c) *Strobe rehearsal*: the ball’s position at one-second intervals leading up to the point it stops in the cup.

1 Introduction

Many people often use digital ways for motor learning through online platforms such as YouTube, social media, and video conferencing software [18, 36]. These new technologies have provided learners with easily accessible instructional content, allowing them to learn the proper movement form by watching and imitating the demonstrated movements. To maximize this effect, *augmented feedback*, providing movement feedback to help learners assess performance across various contexts, has been proposed in the field of HCI to provide learners with additional information beyond what they can normally perceive on their own, facilitating more accurate movement execution and skill development [19, 20, 47, 49, 51, 54]. However, research indicates that these learning approaches, which rely on outwardly observable features such as movement form, do not enable learners to acquire the crucial internal representations or the expert intentions and thought processes that underpin the motor skill acquisition [10, 17, 55].

Mental rehearsal, defined as “the cognitive rehearsal of a task prior to performance” [9], is one of the internal representation and cognitive thought process including intentions or strategies. It can help learners develop a deeper cognitive understanding of motor tasks [9, 11], as it has been argued to be able to pre-activate neural circuits [7, 17, 26, 42]. Despite these benefits, few studies focus on the mental rehearsal as a learning tool for novices because novices typically struggle to form vivid mental imagery of unfamiliar skills, limiting the potential benefits of mental rehearsal [5, 16, 21, 52]. Therefore, to support the motor learning of novices, we explored the visualization of the mental rehearsals of experts, as we now have Extended Reality (XR) that can present visual information directly in front of the learner’s eyes [37, 45].

To explore this concept, we propose *rehearsal reality*, an XR-based application that overlays the mental rehearsals of experts onto their own practice environment. Herein, we focused on golf, a sport in which mental rehearsal is widely practiced [3], and developed *rehearsal reality* for golf putting, based on our insights from formative interviews with expert golfers about the mental rehearsal prior to their putts. We conducted a qualitative study

to understand novices’ subjective learning experiences, which are crucial for motor learning [56], by comparing the *rehearsal reality* experience with first-person videos and a captured video version of *rehearsal reality*.

Our qualitative study with 36 novices revealed three key insights. First, *rehearsal reality* can enhance motor skill acquisition, including force control and body movement. Second, first-person videos of successful putting emphasized internal bodily information (e.g., eye gaze, swing amplitude), whereas *rehearsal reality* highlighted external factors (e.g., distance perception, ball trajectory, timing, speed). Third, XR-based approaches may be more suitable than video-based methods for visualizing mental rehearsals.

This paper contributes to, and extends, current HCI work on motor skill acquisition (e.g., [25, 41, 44, 46, 50, 54]) with a focus on a different domain (golf putting) and a different mental imagery (mental rehearsal), and on cognitive understandings [6] with a novel focus and findings on visualizing the mental rehearsals of experts to enhance motor learning.

2 Rehearsal Reality

To prototype the *rehearsal reality* concept in golf putting, we interviewed six male expert golfers (three certified professional and three teaching professional, mean age: 33.17 years) via video conferencing, investigating how they mentally rehearse putting and identifying which aspects of their mental rehearsal would be suitable for visualization. Drawing on Personal Construct Psychology (PCP) [30], a theoretical framework widely used to explore psychological states in sports, and its retrospective interview protocol [15], we developed our interview questions to investigate the mental rehearsal strategies of expert golfers, ensuring alignment with THE core principles of PCP (see Supplementary Material). We recorded, transcribed, and analyzed these interviews using reflexive thematic analysis [4]. Our analysis revealed three main types of mental rehearsal: *line*, *rolling*, and *strobe*. We developed *rehearsal reality* for golf putting by visualizing these three mental images in an XR environment so that learners can overlay them onto their own practice environment (Figure 2 (a), (b), (c)). It was developed using Unity

along with the Meta XR Core SDK and Meta XR Interaction SDK, and deployed it on a Meta Quest 3 HMD. The following sections describe the specific design of each visualization.

2.1 Line Rehearsal

Line rehearsal visualizes a line from the ball to the cup (Figure 2 (a)). This overlay shows an ideal line in the real world, anchored at the physical ball and the cup via the Quest 3 controllers. Based on our interviews, we chose a faint white color for the line and a width equal to the diameter of a golf ball (42.67 mm).

2.2 Rolling Rehearsal

Rolling rehearsal visualizes an ideal rolling trajectory of the ball from its current position to the cup (Figure 2 (b)). By placing virtual anchors at the ball and the cup with the Quest 3 controllers, users can view a virtual ball rolling from the real ball's position and coming to rest exactly at the cup. Two seconds after stopping, it respawns at the start position and begins rolling again.

We used the physics engine of Unity to simulate the rolling motion of the ball. When the speed of the ball is sufficiently low, rolling friction is constant and air resistance is negligible [53]. Given friction coefficient μ , mass of the ball m , gravitational acceleration g , and known distance d to the cup, the initial velocity, v_0 , that stops exactly at the cup is derived from energy conservation

$$\frac{1}{2}mv_0^2 = \int_0^d \mu mg dx \quad (1)$$

We obtain v_0 and position $x(t)$ at time t as follows:

$$v_0 = \sqrt{2\mu gd} \quad (2)$$

$$x(t) = v_0 t - \frac{1}{2}\mu g t^2 \quad (3)$$

In this study, $\mu = 0.05$ for the practice mat, and $g = 9.8 \text{ m/s}^2$. This setup recreates a trajectory in which the ball rolls from the starting position and stops precisely at the cup.

2.3 Strobe Rehearsal

Strobe rehearsal visualizes the ball's position at one-second intervals leading up to the point it stops in the cup, with the spacing between each position shrinking as the ball nears the cup. (Figure 2 (c)). By anchoring the ball and cup positions with the Quest 3 controllers, the user sees a series of strobe-like snapshots overlaid in the real world, indicating the ball's location at each second. The one-second increments are calculated using the same equations from the *rolling rehearsal* approach.

3 Qualitative Evaluation of the Impact of Rehearsal Reality on Motor Learning

To explore how *rehearsal reality* influences putting learning among novices, we compared the *rehearsal reality* experience with first-person videos and a captured video version of *rehearsal reality*, conducted a four-day qualitative study using a between-participants design, aiming to observe the effects of consecutive days of learning. We adopted a retention-design [43], commonly used in motor

learning studies, and each day's session was spaced approximately 24 h apart.

3.1 Conditions

We prepared three conditions. Each participant was assigned to one of the three and performed the four-day protocol under that assigned condition. Participants only received training on the first three days; there was no training on the fourth day.

3.1.1 Positive Video (PV) Condition. In the **Positive Video (PV) Condition**, participants performed practice swings and also watched a successful putting video (shot by an experienced golfer) from a first-person perspective (1PP) on an external display. This condition was inspired by a prior research indicating that viewing a successful performance can enhance mental rehearsal [2]. We introduced this condition to compare the effects of conventional first-person videos with our experts' mental rehearsal visualizations. The same successful putting video was shown on each of the three training days.

3.1.2 Video Mental Rehearsal (VMR) Condition. In the **Video Mental Rehearsal (VMR) Condition**, participants performed practice swings and also watched a video capturing the *rehearsal reality* experience displayed on an external monitor. Each day, they watched one of the three videos (Figure 2 (a), (b), (c)), giving them the opportunity to see all three applications over the course of three days (the order was counterbalanced). This condition was introduced to differentiate the learning effects of *rehearsal reality* as video vs. in XR.

3.1.3 XR Mental Rehearsal (XMR) Condition. In the **XR Mental Rehearsal (XMR) Condition**, participants performed practice swings and concurrently used the three *rehearsal reality* (Figure 2 (a), (b), (c)) over the course of three days (the order of use was counterbalanced).

3.2 Participants

We recruited 12 participants for each condition, for a total of 36 participants (15 males, 21 females, $M = 25.10$, $SD = 9.01$), all right-handed with no regular golf experience. Five had no prior sports experience; the others had participated in sports such as track and field, tennis, or swimming.

3.3 Procedure

Upon arriving at the lab, participants received a user study briefing, provided informed consent, and completed a basic questionnaire (gender, age, golf experience, sports history). Next, they performed a short warm-up, received a brief explanation of the standard putting grip [1, 39] from the experimenter, took three trial putts, and then performed the 20-ball putting. After a one-minute rest, participants then completed their five-minute training according to their assigned condition. No actual golf balls were struck during this period and participants were only allowed to take practice swings and watch their assigned videos or XR. They could freely move along the practice mat to gauge distance to the cup. A tablet displaying a countdown timer remained visible to participants. When five minutes elapsed, an alarm sounded, signaling the end of the training task. After a one-minute rest, they performed the 20-ball putting

again, followed by an interview. Finally, they were dismissed and reminded of their next scheduled session.

The second and third days followed a similar protocol; however, the participants did not fill out consent forms or the basic questionnaire again, nor did they receive the same grip explanation. On the fourth day, participants performed a warm-up, took three trial putts, and then completed a single 20-ball putting and final interview, with no second 20-ball putting or training that day. All participants then received approximately 30 dollars as compensation and exited the study. Sessions were conducted 24–48 h apart ($M = 33$ h 10 min, $SD = 14$ h 33 min).

3.4 Apparatus

We used a putting mat measuring $0.9\text{ m} \times 5\text{ m}$. Participants used a standard, white R&A approved golf ball. The golf cup was 0.2 mm thick with an inner diameter of 10.8 cm and an outer diameter of 12.8 cm, matching official golf cup dimensions. A hole was cut out for the inner diameter, and the cup was fabricated with a 3D printer. We marked the ball's start position and the cup location 3 m away using chalk on the mat. A blade-type putter club was provided.

3.5 Evaluation

We conducted **semi-structured interviews** to investigate the effect of viewing successful putting videos or the mental rehearsal of experts on the motor learning of novices. Immediately after second 20-ball putting, we asked the participants how they interpreted and leveraged the videos, whether the videos helped or hindered their practice, and what effect they felt the videos had on their performance. To evaluate any carry-over effect, we also asked whether the videos from the previous day influenced the next day's 20-ball putting. The first author conducted the interviews in person, recording the audio with a smartphone app. Interview questions were based on prior research on video-based motor learning support (see Supplementary Material) [44].

3.6 Analysis

We transcribed and analyzed all interview data through reflexive thematic analysis [4], which allows for theoretical flexibility and is widely used in qualitative research. Both inductive and deductive coding approaches were employed, led by the first author to generate and refine themes.

4 Results

We conducted semi-structured interviews with 36 participants: 12 in the PV condition (**P201–P212**), 12 in the VMR condition (**P301–P312**), and 12 in the XMR condition (**P401–P412**). We then conducted a reflexive thematic analysis on the interview transcripts. Table 1 provides an overview of the resulting codes and themes. We report on the impact of video- and XR-based training on motor learning below, focusing on two main themes: *physical* and *cognitive* changes.

4.1 Physical Change

Overall, reports of improvements in motor skills, such as force control and performance, were more common in the VMR and XMR conditions, and less frequent in the PV condition.

Three participants in PV, five in VMR, and six in XMR raised the importance of **force control** during the swing. Among them, two participants in PV, four in VMR, and all six in XMR mentioned that they learned how to regulate their putting strength: “*I figured out roughly how hard to hit the ball.*” (P402).

One participant in PV, seven in VMR, and seven in XMR mentioned a **performance change**. Among them, the one participant in PV (P207) said they managed to “*reduce the ball's impact,*” whereas five in VMR and five in XMR noted improved stability. One VMR participant (P308) and one XMR participant (P401) even perceived changes in their physical movements. P308 explained, “*When I formed a mental image, I did better for some reason,*” whereas P401 said, “*I think my body somehow aligned itself to hit the ball straight.*” One VMR participant and one XMR participant stated that it was not particularly helpful.

4.2 Cognitive Change

Overall, participants in the PV condition more frequently reported gaining a cognitive understanding of their movement form, whereas those in the VMR and XMR conditions more often mentioned focusing on the external environment.

Only five participants in the PV condition reported learning about the **gaze** of successful putters, with one participant noting, “*It was helpful in facilitating understanding regarding the position of one's gaze*” (P212). Furthermore, five in the PV condition reported understanding of the **swing amplitude**, as “*It was really easy to understand because I could compare how my own practice swing differed from it*” (P204).

Six participants, three in the PV and three in the VMR conditions, made comments regarding their perception of **distance**. Among them, two PV and two VMR participants said it was easier to gauge distance, whereas one from each condition found it difficult. Additionally, five and eight participants in the VMR and XMR conditions, respectively, discussed understanding **ball trajectory and angle**. Of those, all eight in the XMR condition and four in VMR reported improved comprehension of the trajectory or angle. Eight and two participants in the VMR and XMR conditions, respectively, said they gained a better sense of **timing** as the ball rolled. One participant in each of those groups (P303 and P407) noted a strategy of “*memorizing the ball's position after one second.*” Furthermore, seven and six participants in the VMR and XMR conditions, respectively, mentioned the **perceived speed** of the ball, with six in VMR and five in XMR stating that they now understood both the initial speed off the putter and the sense of deceleration: “*I didn't understand anything, so the sense of speed and correctness became a vital reference—like a model example.*” (P303).

Ten VMR participants and seven XMR participants also reported a **change in their mental focus**. P303 noted that “*because I wasn't over-focusing on my hands ... something automatically did the adjustment for me.*” Additionally, one participant in PV, three in VMR, and four in XMR stated that they were **able to predict** whether the ball would go in the cup immediately after striking it. P303 explained, “*I can predict if my shot will enter the goal,*” and P402 said, “*The moment I hit the ball, I can tell whether it will be successful or not.*”

Table 1: Overview of reflexive thematic analysis results. Numbers in parentheses indicate the number of participants (maximum 12) who mentioned that topic.

Code	PV Condition	VMR Condition	XMR Condition
Force Control	Understood (2), Did not understand (1)	Understood (4), Did not understand (1)	Understood (6)
Performance Changes	Reduced impact (1)	More stable (5), Changed body movement (1), Not helpful (1)	More stable (5), Changed body movement (1), Not helpful (1)
Gaze	Understood (5)	No mentions	No mentions
Swing Amplitude	Understood (5)	No mentions	No mentions
Distance Perception	Easy to gauge (2), Hard to gauge (1)	Easy to gauge (2), Hard to gauge (1)	No mentions
Trajectory / Angle	No mentions	Improved understanding (4), Not helpful (1)	Improved understanding (8)
Timing	No mentions	Could grasp timing (8)	Could grasp timing (2)
Perceived Speed	No mentions	Understood initial speed and deceleration (6), Did not understand (1)	Understood initial speed and deceleration (5), Did not understand (1)
Change in Mental Focus	No mentions	Changed (10)	Changed (7)
Predicting Ball Outcome	Became able to do so (1)	Became able to do so (3)	Became able to do so (4)
Information Quantity	No mentions	Low (3), Appropriate (1), High (5)	No mentions

Six participants in the VMR condition reported on **the amount of information** provided by the videos. Among them, three pointed out the lack of information in the *line rehearsal*. Four noted the abundance of information in the *rolling rehearsal*. Furthermore, regarding the *strobo rehearsal*, one participant felt it offered a great deal of information, whereas another described it as falling somewhere between the line and *rolling* rehearsals.

5 Discussions

5.1 Motor Skill Acquisition

Our results indicate that *rehearsal reality* can enhance motor skill acquisitions such as force control and body movement. These results align with findings in prior HCI work on visual mental imagery featuring other practices, e.g., dancing practices [25, 41, 46, 50] and strength training [44, 54]. Our work serves to extend these findings to a different domain (golf putting) and a different mental imagery (mental rehearsal). Dancing and strength training share a specific aspect with golf putting: they all foreground learning particular ways of moving, yet, golf involves the use of equipment, introducing a level of difficulty that distinguishes it from dancing and strength training. In Gentile's Taxonomy [13], a systematic classification system to categorize motor skills and movement, whether an object is manipulated is a key factor influencing task difficulty, with object manipulation being more challenging than moving the body alone. As such, our visual cues for golf putting tended to be more difficult than those in dance practices and strength training in prior work [25, 41, 46, 47, 50, 54]. Our work shows the potential of *rehearsal reality* in enhancing motor learning in a specific type of technology: 1PP videos and XR.

5.2 Cognitive Understandings

Our results also indicate that standard 1PP video emphasizes on internal bodily information like motor form, e.g., eye gaze and club swing amplitude, whereas *rehearsal reality* tends to highlight external aspects, such as distance perception, trajectory, timing, and speed, when the ball rolled, as well as an ability to predict whether the ball would go in after the stroke. As such, a subjective 1PP, reflecting an expert's internal viewpoint, may offer stronger external cognitive insights than a mere objective (standard) 1PP. This finding adds to, and extends, prior work [6] suggesting that 1PP video [12, 14, 22–24, 28, 34, 48] benefits cognitive task learning, whereas 3PP video [8, 27, 29, 31–33, 35] benefits motor form acquisition, by introducing the perspective of whether subjective information is incorporated in 1PP videos. Consequently, one could posit the following hierarchy for internal cognitive factor learning: 3PP videos, objective 1PP, and subjective 1PP. Conversely, for external cognitive factor learning, the order might be reversed: subjective 1PP, objective 1PP, and 3PP. This hierarchy serves as a useful guideline for instructors and learners in selecting the most appropriate video perspective to match their skill acquisition goals.

5.3 Video vs XR

Reports from the VMR condition indicated that *rolling* and *strobo rehearsals* contained a considerable amount of information, whereas such reports were not obtained from the XMR condition. This suggests that XR may be more suitable than videos for visualizing mental rehearsal, as it reduces cognitive load more effectively. This finding aligns with those reported by prior work [38, 40] and highlights the temporal and spatial integration characteristics of XR as key contributors to reducing cognitive load. We extend this

prior work to a different context: the visualization of the mental rehearsals of experts, demonstrating XR as a suitable tool for helping novices understand these rehearsals.

5.4 limitation

All participants in this study were golf novices, hence, we cannot simply apply our findings to intermediate or advanced players. The novices can easily accept the *rehearsal reality*, as “*I didn’t understand anything, so the sense of speed and correctness became a vital reference—like a model example*” (P303), whereas intermediate or advanced players may already have their own mental representations, leading to conflict with the existing mental imagery [44, 46].

Further, because our main goal here is to understand novices’ subjective learning experiences with rehearsal reality, we do not discuss improvements in putting performance at this stage. Therefore, a quantitative study should be conducted in the next phase to investigate rehearsal reality’s contribution to motor learning.

6 Conclusion

In this study, we propose the concept of *rehearsal reality*, an XR-based application that overlays the mental rehearsals of experts onto the real-world environment, and investigated how it affects the motor learning of novices in putting. Our findings suggest that the first-person visualization of mental rehearsals could enhance motor skill acquisitions such as force control and body movement, and may deepen external cognitive aspects compared to conventional first-person videos. Furthermore, our study indicates that XR may be more suitable than videos for visualizing mental rehearsals. These findings shed light on several design strategies and challenges associated with the usage of the mental rehearsal of experts as visual cues in motor learning with video and XR. For future studies, we plan to conduct a quantitative study to evaluate whether the putting skill of novices can be improved through the visualization of the mental rehearsals of experts.

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